

A novel framework for artificial intelligence explainability via the Technology Acceptance Model and Rapid Estimate of Adult Literacy in Medicine using machine learning

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ABSTRACT

The significant proliferation of AI-empowered systems and machine learning (ML) across various examined domains underscores the vital necessity for comprehensive and customised explainability frameworks to lead to usable and trustworthy systems. Especially in the medical domain, where validation of methodologies and outcomes is as important as the adoption rate of such systems, the requirements of the depth and the level of abstraction of the explainability are particularly important and necessitate a systemic approach to ensure a proper definition. Explainability and interpretability are important usability and trustworthiness properties of AI-empowered systems and, as such, constitute important factors for technology acceptance. In this paper, we propose a novel framework for explainability requirements in AI-empowered systems using the Technology Acceptance Model (TAM).

This framework employs targeted ML (hierarchical clustering, k-means or other) to acquire a user model for personalised, multi-layered explainability. Our novel framework integrates a rule-based system, which guides the degree of trustworthiness to be achieved based on user perception and AI literacy level. We test this methodology in the case of AI-empowered medical systems to (1) assess and quantify the doctors' abilities and familiarisation with technology and AI, (2) generate layers of personalised explainability based on user ability and user needs in terms of trustworthiness and (3) provide the necessary environment for transparency and validation. To assess and quantify the doctors' abilities we have considered Rapid Estimate of Adult Literacy in Medicine (REALM) a tool commonly used in the medical domain to bridge the communication gap between patients and doctors.

1. Introduction

Modern medicine studies biological markers (biomarker) to assess health conditions (Poste, 2011) and improve the patient's journey. Modern technological advancements, allow smart wearables to measure various biometrics like heart rate variability, body temperature and menstrual cycles, and offer recommendations to its users. This results in a heightened perception of how the body works and changes overtime and how it depends on external variables like nutrition and activity. In many cases, the physiological data collected are analysed via Artificial Intelligence and machine learning.

At the same time, digital tools of artificial intelligence (AI) and big data analysis demonstrate strong potential to improve symptom identification and to forecast treatment outcome. Particularly promising is the use of machine learning-based analysis, which attempts to imitate the way the human mind trains and operates. Specifically, machine learning (ML) is an AI discipline referring to the way a computer can

use algorithms and statistical analysis to recognise patterns in various data and from those extract information and automate medical decision support. The necessity of transparency in various fields can differ, yet it becomes increasingly vital as the data involved becomes more sensitive and decisions influenced by AI carry more potential for serious consequences.

Thus, AI is the driving force that can bridge user needs and develop tools, the analysis of health-related data and measured body signals can create new medical pathways or improve on existing ones. It can also enhance the quality of life and extend life as a time-sensitive variable. However, problems of relevance and misuse arise, as per the outcomes of AI-empowered systems that use ML algorithms. Validation difficulties emerge either in the context of confirmation bias or bias induced from the stakeholders responsible for validation and administration. Patients' health and data safety requires the establishment of frameworks that ensure all underlying technologies are adequately elucidated

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and interpreted. As complexity of medical recommendation is very high, related interventions hold significant risks and potential threat to life and well being must be taken into serious account. The necessity of transparency in various fields can differ, yet it becomes increasingly vital as the data involved becomes more sensitive and decisions influenced by AI carry more potential for serious consequences.

In the last decade, the field of Explainable AI (XAI) has emerged and evolved (Biecek & Burzykowski, 2021; Holzinger et al., 2022; Vilone & Longo, 2020; Virvou, 2022). XAI offer the necessary context for doctors and patients to use AI-powered machines safely, transparently, and reliably. Systems empowered with artificial intelligence are considered intelligent, because they can imitate human behaviour in predicting and proposing interventions and recommendations that can improve real life outcomes. The system capabilities are measured on their communication abilities, internal knowledge, intentionality, and creativity (Schank, 1987). More recently, with the rise of ML models that are facilitated by strong language processing models (Large language models, LLMs) systems, such as ChatGPT, are further measured on their reasoning, speed and conciseness (OpenAI, 2023).

Whatever the measure may be, transparency is an important factor, especially so in sensitive domains, like the medical domain. The validation of such systems by its stakeholders, users, doctors and regulators needs to be very precise and on point, in order to avert risks and complications.

Explainable AI (XAI) is a commonly used term of all those measures that include graphical and textual descriptions of how a trained model classifies input and makes decisions. An AI-empowered system that is not explained is considered opaque. Therefore, system transparency and explainable decision making should be considered a given and the demand for such systems a prerequisite. An important design parameter that should be considered with regard to explainability in AI-empowered systems is the users' ability and familiarity with such technologies (Miller, 2019). Thus, a combination of scientific language and simpler approaches should be contemplated (Arrieta et al., 2020; Ngo, Kunkel, & Ziegler, 2020; Tsai & Carroll, 2022). Responsible AI (RAI) extends the concept of XAI further and refers to the deployment of AI-empowered systems where transparency and alignment with the basic human values of ethics and accountability is of most importance (Floridi et al., 2018).

In this paper, we engage into the creation of a novel framework for personalised explainability requirements in AI-empowered medical applications by merging machine learning and innovation theories. In particular, we have created a system that combines clustering with a rule-based decision system, which are embedded in a methodology aimed at promoting AI transparency. The rule-based systems utilises the Technology Acceptance Model (TAM) for evaluation of user behaviour with regard to technology adoption and clustering via quantification of user's perspective on usefulness and ease of use. TAM, which was originally introduced by Fred Davis, offers a good basis to be used to increase trustworthiness and transparency in the new AI methods. TAM is a dominant framework that has been extensively used for investigating the various factors that influence users' acceptance of technology. TAM postulates that the mediating role of two pivotal variables, namely perceived ease of use and perceived usefulness, is integral to the complex relationship between external system characteristics and the potential adoption and usage of the system. Within this, targeted machine learning techniques, such as hierarchical clustering, k-means, or other algorithms, are used to construct a model that facilitates personalised and multi-layered explainability. To achieve this, we consider two crucial external variables: the users' AI literacy level and the level of abstraction in the explanation provided. These variables play a significant role in measuring users' perceptions of usefulness and their perceived ease of use when interacting with AI-empowered systems. Our innovative framework integrates machine learning with a rule-based system, which allows us to determine the appropriate degree of trustworthiness based on the users' perceived usefulness and ease

of use, as specified by TAM. This integration ensures that the level of trustworthiness in the explanations is tailored to each user's specific needs and abilities.

Moreover, in this paper a comprehensive implementation strategy is outlined, which centres around personalisation and is derived from targeted clustering techniques based on the principles of TAM. By employing targeted clustering, we create tailored explanations for users based on their unique characteristics and requirements. To validate the effectiveness of our methodology, we conducted a case study using AI-empowered medical applications. The objectives of this study were threefold:

1. Assess and quantify the perspectives of doctors regarding the relevant parameters, such as AI literacy and the level of abstraction in explanations.
2. Generate layers of personalised explainability based on individual user abilities and needs in terms of trustworthiness, ensuring that the explanations are tailored to the users' specific requirements.
3. Provide a transparent environment that fosters understanding and validation of the AI-empowered medical applications' decisions, increasing overall trust in the system.

To gauge patient understanding and ability, tools like REALM (Rapid Estimate of Adult Literacy in Medicine), have been extensively used in the medical domain. Similarly, equivalent tools are employed in other domains to bridge the communication gap between experts and users by assessing comprehension. Even though, we do not directly use REALM in our framework, we considered a similar approach for setting the maximum XAI layers, i.e. clusters.

By implementing our framework and testing it in the context of AI-empowered medical applications, we aim to establish a robust foundation for delivering personalised, trustworthy, and transparent AI systems in various domains. We evaluate this methodology within the context of the medical field. Initially, we exhibit the findings from a survey we conducted examining the extent of AI-empowered applications' adoption among medical professionals. Subsequently, we employ this innovative approach to dissect the requirements gathered through questionnaires, leading to the formulation of a transparency and validation framework that aligns with these requirements. In our previous related research, we have outlined and developed an AI-empowered application called NuhealthSoft, that uses neural networks and analytical tools to assess biometrics and identify health states. No XAI tools were implemented in the system, previously. Instead, we used NuhealthSoft to deploy the framework introduced in this study.

This paper is structured as follows: Section 2 focuses on elucidating the foundational human behavioural theories we leverage and reviewing relevant past studies. In Section 3, these foundational theories guide the development of an innovative framework, that incorporates the definition of AI literacy requirements and the level of required explainability abstraction. We refer to this framework as PINXEL (Personalised INterface for eXplainability baSEd on Levels) 2, which signifies transparency and validation for AI-empowered applications. This methodology delineates and expands multiple layers of explainability and transparency, integrating the clustering methodology shown in Fig. 8 and elaborated in Section 4. In Section 4, we use PINXEL to analyse a questionnaire-based survey and define the personalisation layers using TAM as outlined in Figs. 2, 8 and 7. The goal is to enhance the adoption rates of AI-empowered applications via a rule-based system, where explainability and transparency are considered important adoption properties. In Section 5 we implement and present those layers using NuhealthSoft as test-bed. Lastly, in Section 6, we provide a summary of the paper and present potential directions for future research.

2. Related work

2.1. Evaluating human behaviour

There are numerous theories that are commonly used to understand, predict and explain human behaviour, in regards to different hypothesis suggested by researchers, and in relation to the adoption of innovative new technologies and products. The most commonly used are:

1. Unified theory of acceptance and use of technology (UTAUT): UTAUT combines elements from eight different models including TAM. It identifies performance expectancy, effort expectancy, social influence and facilitating conditions (Jacob, Grosse, Morana, & König, 2023; Venkatesh, Morris, Davis, & Davis, 2003).
2. Diffusion of Innovation (DOI): According to Rodgers, the diffusion of an innovation is highly correlated with how a population perceives a proposed idea or product. If the proposed idea or product is considered *innovative*, diffusion is probable and, thus, its adoption will increase accordingly (Rogers, 2010).
3. Bass Diffusion Model: To outline the process of how new products get adopted in a population a mathematical model was developed. This model distinguishes adopters as innovators or imitators (Kumar, Guha, & Chakraborty, 2022; Norton & Bass, 1987).
4. Theory of Planned behaviour (TPB): TPB suggests that individual intention is determined by attitudes, subjective norms and perceived behavioural control (Ajzen, 1991; Yuriev, Dahmen, Paillé, Boiral, & Guillaumie, 2020).
5. Critical Mass Theory: This theory suggests that innovations become self-sustaining and grow rapidly once a critical mass of adopters is gained (Dobija, Hryckiewicz, Zaman, & Puławska, 2022; Markus, 1987).
6. Social Cognitive Theory: This theory emphasises on the importance of social influence and observational learning in the adoption process (Bandura, 1999; Beauchamp, Crawford, & Jackson, 2019).
7. Consumer Decision Process Model: This model focuses on the different stages of a consumer decision prior to adoption or purchase (Bangsa & Schlegelmilch, 2020; Cox & Rich, 1964).

In this paper, we chose the Technology Acceptance Model (TAM), which is introduced in the following subsection, because of its comprehensibility and wide acceptability. More importantly, its simplicity allows for adapting it in a digital pipeline, where decisions can be automated. External variables are also commonly associated with TAMs defining parameters.

2.1.1. Technology acceptance model

According to the Technology Acceptance Model (TAM) (Ajzen, 1991; Davis, 1989), the acceptance of new technological systems depends on two external factors which are considered as measurements of *intention to use* and *likelihood of adoption*. The users' perception of value extraction from that technology is defined as *perceived usefulness*. The level of effort as a usability parameter in TAM is defined as *perceived ease of use*.

TAM has been extensively applied within the medical field to assess user acceptance and integration of emerging technologies, such as electronic health records, tele-medicine, and generic health information systems. In Chau and Hu (2002), the investigation into the acceptance of telemedicine by healthcare professionals concludes that perceived ease of use and perceived usefulness are pivotal factors in determining the likelihood of technology adoption. In Holden and Karsh (2010), the authors consider the application of the TAM as a guiding principle for designing and developing healthcare applications. Furthermore, in Tung, Chang, and Chou (2008), the researchers augment the TAM by integrating it with the Diffusion of Innovations (DOI) theory, to scrutinise the adoption of electronic logistics information systems within hospital information systems.

2.1.2. Informational communication gap in the medical domain

The informational communication gap between doctors and patients is an important factor of responsible and inclusive health care. The domain offers a variety of tools for addressing this problem. This particular gap refers to the use and understanding of medical terminologies, acronyms, or complicated diagnoses (Gallagher & Updegraff, 2012; Parker, Ratzan, & Lurie, 2003). REALM (Rapid Estimate of Adult Literacy in Medicine) is a tool used to evaluate an individual's capability to read and articulate ordinary medical language and simple terms for anatomy and diseases (Murphy, Davis, Long, Jackson, & Decker, 1993). TOFHLA (Test of Functional Health Literacy in Adults) is used to assess a patient's comprehension and deciphering of directives, like those typically found on prescription drug labels (Parker, Baker, Williams, & Nurss, 1995). Newest Vital Sign (NVS) is a tool that scores a patient's reading proficiency and numerical abilities using a nutrition label from an ice cream package (Weiss et al., 2005). Though, we do not use these tools as is, in our framework, they are instead considered as a basis for setting a maximum number of distinct groups with varying understandings of AI-related themes. Especially REALM was evaluated and adapted for the purpose of this research, and the development of the proposed framework.

2.2. Technology and applications

In our research we have outlined an application using micro-services (Panagoulas, Virvou, & Tsihrintzis, 2023a), that uses ML-based models to analyse health related data to make classifications, predictions and recommendations. Specifically, we obtained high precision in predicting crucial health states by employing features derived from blood tests. This enabled us to predict weight categories with an accuracy of 85% (Panagoulas, Sotiropoulos, & Tsihrintzis, 2021a, 2021b, 2021c, 2022b). Similarly, these features facilitated predictions of metabolic syndrome with an accuracy of 86% and high systolic blood pressure with an accuracy of 74%. We have integrated these methodologies into a system capable of analysing blood exam data to assess health conditions, provide guidance, and automate health interventions (Panagoulas, Virvou, & Tsihrintzis, 2023b). Regulatory considerations, vital for health application development, were also addressed in Panagoulas, Virvou, and Tsihrintzis (2023c). We have also used extreme value analysis in dietary intake and thus draw important conclusions (Panagoulas, Sotiropoulos, & Tsihrintzis, 2022a).

2.2.1. NuhealthSoft

In this paper we use NuhealthSoft as a test-bed to evaluate PINXEL (introduced in detail in the following section) and also to add personalised XAI functionality to the system. NuhealthSoft uses ML-based models and nutritional and biochemical input data to classify users with regard to a series of medical concerns, including blood pressure, weight and metabolic syndrome (Panagoulas et al., 2021a, 2021b, 2021c, 2022b, 2023c). Moreover, NuhealthSoft (Fig. 1) offers nutritional suggestions tailored to a user's specific needs and blood works.

Clearly, doctors and regulators are necessary validators in the development process of NuhealthSoft. This has been the main incentive for this research. For that reason, we have looked into the specific requirements of the doctors to ensure a smooth validation process, but also provide a transparent framework for the actual users of the system.

3. Overview of PINXEL

In this study, we propose, outline and implement a novel framework named PINXEL. The acronym stands for Personalised **I**nterface for e**X**plainability basEd on Levels. The levels we have considered are those of AI explainability and abstraction. PINXEL uses TAM to assess the characteristics and abilities of the users. Those characteristics define different clusters and for each cluster a unique layer of explainability is provided, thus a personalisation approach is followed.

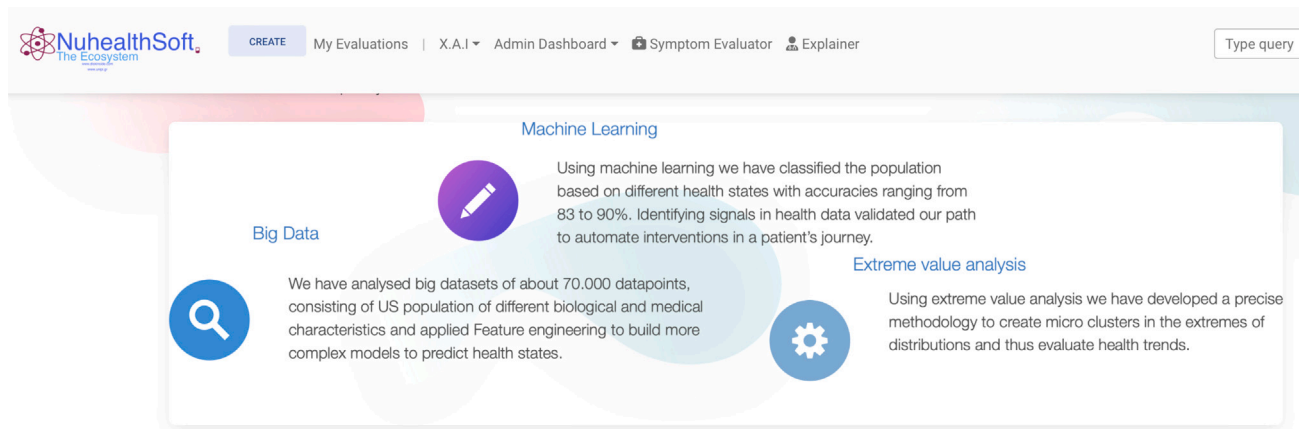


Fig. 1. Screenshot from NuhealthSoft.

Technological literacy encompasses the competency to comprehend, utilise, and critically assess technology across diverse contexts, coupled with the capacity to participate in technology-related decision-making processes. This literacy spans an array of knowledge, abilities, and dispositions that equip individuals to adjust and flourish in a swiftly evolving technological space. Frequently, technological literacy is deemed a vital element of contemporary education and readiness for the workforce.

Based on existing research (Ng, Leung, Chu, & Qiao, 2021), different categories exist that delineate varying degrees of **AI understanding**. This begins with a user's capacity to differentiate between technologies that employ artificial intelligence and those that do not. More sophisticated users can analyse data, identify the sensors responsible for gathering it, and comprehend the ways in which a system interacts with its environment to offer suggestions and forecasts. In general, understanding and ability to evaluate AI is more commonly referred to as **AI literacy**.

The **level of abstraction** pertains to the extent and intensity to which intricate concepts are highlighted and conveyed. In the area of ML, varying levels of abstraction should be adopted to accommodate diverse audiences and objectives. A higher level of abstraction is employed to portray a concept in a simpler fashion, while a lower level of abstraction includes more complex terminologies that are more appropriate for audiences possessing expertise and domain-specific knowledge (Garfinkel, 2019; Marr, 1982; Mitchell, 1997).

As stated in the previous, the two parameters that delineate technology acceptance as per the TAM are perceived usefulness and perceived ease of use. In the following, we associate these two parameters with AI literacy and the level of abstraction to provide personalisation of XAI. A greater comprehension of potential advantages and applications of a technology heightens its likelihood of being perceived as beneficial. In Teo (2011), the association between computer *self-efficacy* and perceived usefulness is investigated. Self-efficacy refers to a person's conviction in their capability to successfully execute a task within a particular domain, an attribute closely tied to the individual's knowledge of that domain — in other words, their literacy.

In Hsu and Lu (2004), the authors argue that an individual's background and experience is related to perceived usefulness, both concepts closely related to ones literacy level of a domain or theme. In Scherer, Siddiq, and Tondeur (2019), the authors agree that perceived usefulness is related to the students' information literacy, in their appreciation of the benefits of e-portfolio systems. Finally in Lin and Tsai (2016), the students of physics with higher learning self-efficacy have a higher perceived usefulness of the learning materials and strategies.

Within the framework of technology adoption, factors such as user experience, usability, and a higher level of abstraction enhance a system's accessibility and comprehensibility and thereby improve the

perceived ease of use of the system. In Karray, Alemzadeh, Saleh, and Arab (2008), the authors provide a comprehensive review of human-computer interactions, emphasising the significance of abstraction in user interface design. As per Cañas and Novak (2012), meaningful learning is linked to concept mapping and abstraction, proposing that the representation of intricate ideas and relationships through abstraction can render them more accessible, subsequently amplifying their perceived ease of use.

3.1. AI literacy levels in PINXEL

In PINXEL we considered three levels of AI literacy, namely basic, average and expert. This is illustrated in Fig. 2. At each level, different themes of AI and ML are acknowledged based on their corresponding complexity, as follows:

- **Basic level:** At this level, general concepts and principles of AI are presented. This involves discussing concepts like ML, neural networks, how AI uses data to make decisions or predictions, the incentives behind creating a smart system, the people involved and affected by such an AI-empowered application, high-level ethical considerations and potential societal impacts of a proposed AI-empowered system.
- **Average level:** Here takes place a more detailed explanation of the specific algorithms and techniques used in the AI-empowered system. This includes details about the model architecture, the data it was trained on, the learning algorithm, the performance metrics and so on.
- **Expert Level:** At this level, the specifics of the model's inner workings takes place. This is where concepts like *global* and *local* XAI are analysed.

This kind of separation into three levels is commensurate with the generally accepted principle in most educational systems of going from the general to the more specific as one advances on the theme he/she is educated on Collins and Halverson (2018), Jarvis (2004). Especially in the medical domain, as we will show in more detail in the following section, tools like REALM are utilised to assess patients' abilities, in order to bridge the communication gap.

According to Chowdhary (2020), effective knowledge representation in AI should be comprehensive, clear, and concise, with minimal irrelevant details, but still allowing access to these when necessary. It must also be fast and computable with standard procedures. The quality of representation largely depends on algorithms, structures, hardware, and knowledge dissemination. Fundamental components of AI representation include lexical (vocabulary), syntactic (sentence structure), semantic, and procedural (creation, modification, and interpretation methods) aspects. In our proposed framework, we combine and restructure these aspects of knowledge representation, to also be domain and application specific, while taking into consideration user ability.

Personalised **IN**terface For **X**plainability basEd On **L**evels

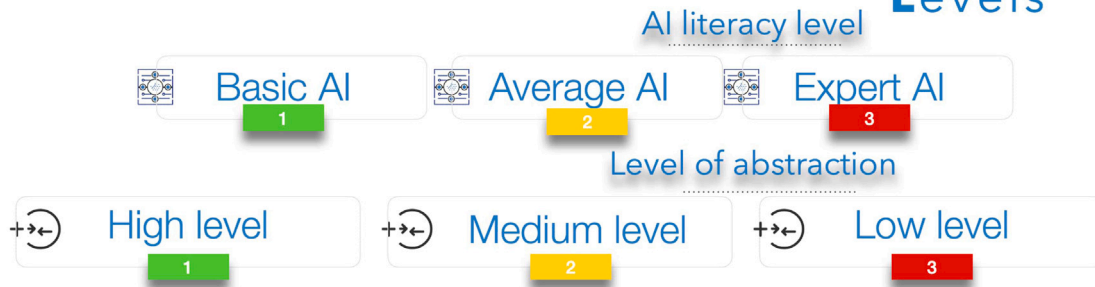


Fig. 2. PINXEL.

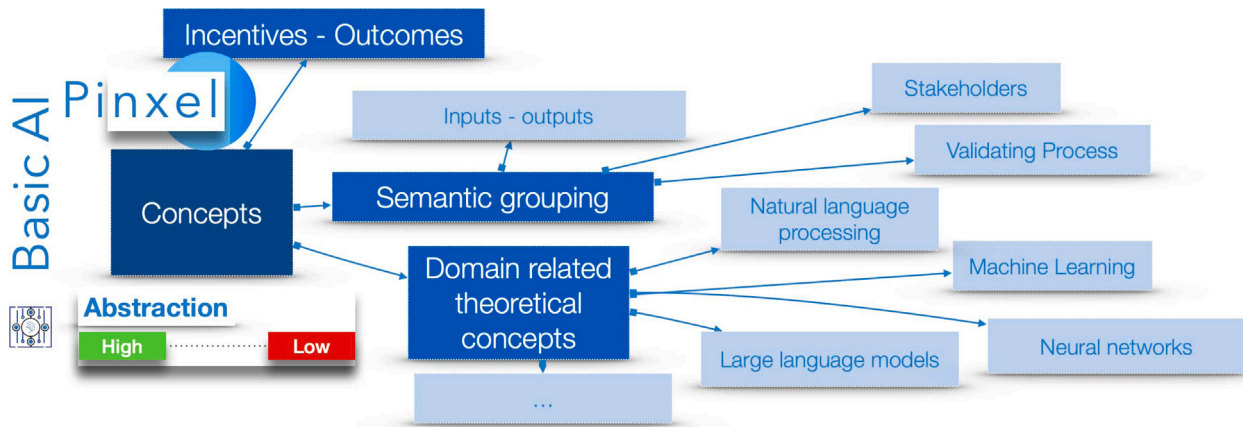


Fig. 3. Basic AI.

3.1.1. Basic level of AI literacy

As can be seen in 3, incentives and outcomes of ML, semantic grouping of key variables and domain related theoretical concepts are used as explanation prompts. *Semantic grouping* in AI is the process of structuring data and concepts based on relationships, resemblance and context (Manning, Raghavan, & Schütze, 2008) in a way that knowledge can be easily understood. For a better definition of the ML process inputs and outputs, stakeholders and processes are grouped and represented semantically.

Domain-related theoretical concepts are grouped themes, designed to target subject areas that are incorporated in AI-empowered systems (Chen et al., 2020). In this area, natural language processing, machine learning, Large language models and other specific subjects, that are commonly utilised in the AI-empowered applications, are subjects, for the user to get familiarised with.

3.1.2. Average level of AI literacy

As can be seen in Fig. 4 a deeper explanation of the model is offered to the users of this category. The “model deconstruction” refers to the architecture of the model, that includes the layers of the model, the activation functions, the weights and biases, and the loss functions. In detail, the “model layers” refer to the hierarchical structure of the artificial neural networks (ANN), where processing of the trained features and knowledge extraction are taking place. The “model layers” can be split into three categories, namely the input, hidden and output layers (Goodfellow, Bengio, & Courville, 2016). The “activation functions” refer to the mathematical functions used to add non-linearity and determine whether a models neuron should be activated or not, depending on the processed input (Haykin, 2009). The “weights and biases” are the adjustable parameters that define the connection density between the neurons of the neural network and the offset amount applied to

the incoming signal. Weights are multipliers added on that signal. The biases are added on the weighted inputs and are continuously updated during the training process to ensure minimisation of error between predicted and actual output (Bishop, 2006). Finally, the loss function (alternatively termed cost or objective function) refers to the quantification of the difference between predicted and actual output also known as ground truth. The minimisation of the loss function is considered a metric of the model’s performance (Murphy, 2012).

The second part of the “model deconstruction” phase is related to the measurement of fairness, bias and dataset size. Fairness and bias refer to the equitable treatment of the different groups included in the dataset used for the training of the model (Barocas, Hardt, & Narayanan, 2018).

In the “model evaluators” phase, metrics and plots are provided with respect to the measured model accuracy. These are the confusion matrix, the classification report and the threshold review. The confusion matrix or error matrix is a table utilised to visualise the performance of a ML algorithm or method, which shows the number of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) for the different classes predicted by the model (Hastie, Tibshirani, & Friedman, 2009). The classification report summarises key performance metrics of the trained model and by association provides an insight of the model’s ability to predict and classify (Alpaydin, 2020). The threshold in probabilities sets the boundary between classes. Usually the boundary is equal to 0.5 but depending on circumstances that boundary can differ (Bishop, 2006). To define the optimum threshold for a classification problem the receiver operating characteristic (R.O.C.) curve and Youden’s index are more commonly utilised.

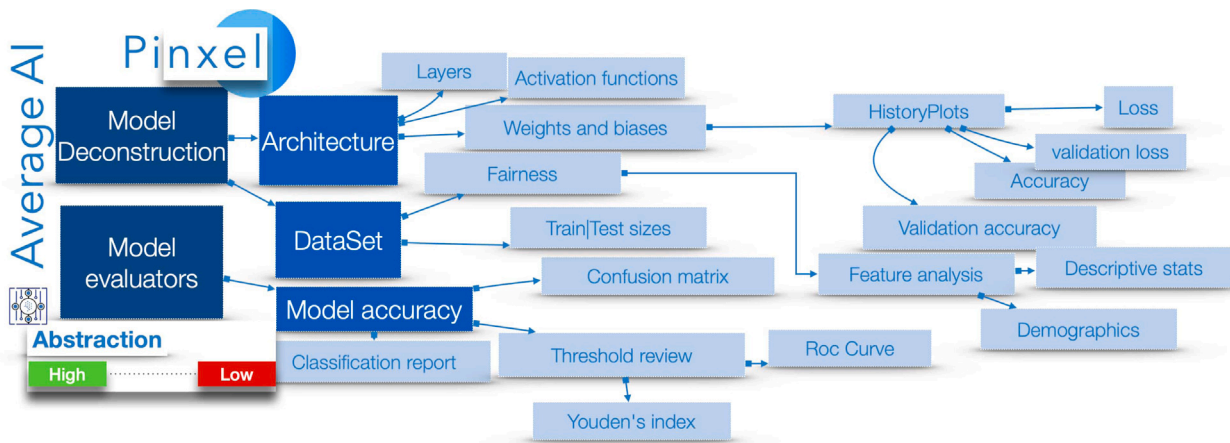


Fig. 4. Average AI.

3.1.3. Expert AI literacy level

For the Expert AI literacy level, the global and local parameters of the trained model are analysed and an analysis is presented of the different parameters leading to a trained and deployed model's decision, as in Fig. 5. "Global AI explainability" should provide an understanding of how the model behaves, considering the input space as a whole and the relationship between input and predictions (output) (Arrieta et al., 2020). *Feature importance* ranks the importance of each input (feature) based on the contribution of the model's performance and predictive ability. The Partial Dependence Plots (PDP) is a visualisation of the relationship between the target(s) and one or more features, while other features are considered constant. A PDP is used to evaluate a model's response to changes of dependent variables. *Accumulated Local Effect plots (ALE)* are similar to PDPs as they average the local changes in predictions by accounting for the relationship between interacting features. *Global surrogate models* approximate the behaviour of a model in review by mimicking its predictive process. Thus, insights are gained of the reviewed model by analysing the behaviour of the surrogate. *Rule extraction* provides a set of easy to understand rules that approximate the ML pipeline and decision process. *Dimensionality reduction* techniques are used to decrease the dimensional space of data used for model training and its relationship with the extracted predictions.

On the other hand, the local explanations show a specific variable contribution on different tested scenarios and, thus, an overall idea of how well the model can perform. Using local explainability, a user can proceed in evaluating the model trained in a more agnostic manner, while the developer can validate the architectural choices made for the building of the model. When examining a variable for explanation, both the variable and the value of the variable are considered. In that scope, a variable can be generally important, but the effect of the variable for the specific evaluated instance may be non-significant (Biecek & Burzykowski, 2021). *Local Interpretable Model-Agnostic Explanations (LIME)* use local coefficients to provide insights on the features' importance for a particular prediction. SHapley Additive exPlanations (SHAP) are again feature importance measurements for individual model decisions based on cooperative game theory, where different possible feature combinations and feature contributions are considered. *Individual Conditional Expectation (ICE) plots* are relationship visualisations of feature and outcomes, when all the other features are considered constant, thus examining different scenarios and feature contribution to a prediction. *Anchor explanations* are explanations based on rules that use anchors or conditions with which the model is more probable to produce an outcome.

Depending on a pre-determined level of abstraction, the explanation technique and depth of the previously discussed themes grouped by AI literacy levels of PINXEL, varies.

3.2. Levels of abstraction in PINXEL

In computing, abstraction often involves managing complexity by hiding lower-level details and providing simplified, high-level interfaces. The more abstract a concept or system, the more it is removed from its low-level details. This is often visualised as a stack of layers, with the lowest (most concrete) level at the bottom and the highest (most abstract) level at the top. This stack can be thought of as having "depth" with more abstract concepts being "deeper" in the sense that they are further removed from the low-level details. Similarly, "shallow" might refer to concepts closer to the low-level implementation. In Fig. 6, the different layers of abstraction are presented, as building blocks of PINXEL.

3.2.1. PINXEL- high level of abstraction

The first explainability layer, which refers to the highest level of abstraction can be seen in Fig. 6. It involves, NLPs for generalising the different concepts, related to the previous section (AI literacy levels). The NLPs are fed with pre-structured prompts, in a pre-defined space determined by literacy level. For example in this abstraction level, the prompt is pre-constructed starting with "Please explain simply" and using the themes provided by the user's specific AI literacy levels (see corresponding Figs. 3, 4, 5). The user can then request further elaboration on the first provided answer, again using a pre-constructed prompt. An implementation example can be seen in Fig. 17.

3.2.2. PINXEL- medium level of abstraction

For the Medium level of abstraction, NLPs with more advanced prompts are employed. Compared to the previous layer, the difference here is that the explanations via the NLPs is more technical. For example, an explanation of feature importance in general in ML may be provided using related literature, as well as an explanation as per the specific feature importance in the domain of this specific application according to the associated literature. Here, we provide the user with more freedom to navigate through the explainability module of the system, using the NLPs with fewer constraints. As can be seen in Fig. 6, a more technical explanation is requested with references included and the answer should be domain specific, i.e related to the models implemented in the application examined. As a second prompt, similarly to the previous level, the user can request for further elaboration on the answer provided, whereas a user-specific point can be provided.

3.2.3. PINXEL- low level of abstraction

Finally, for the Low level of abstraction, the user is presented with tables and graphical representations of the properties, attributes, and results of various machine learning models. The graphs and tables are those accustomed in the domain of XAI and examples of those can be seen in the following sections.

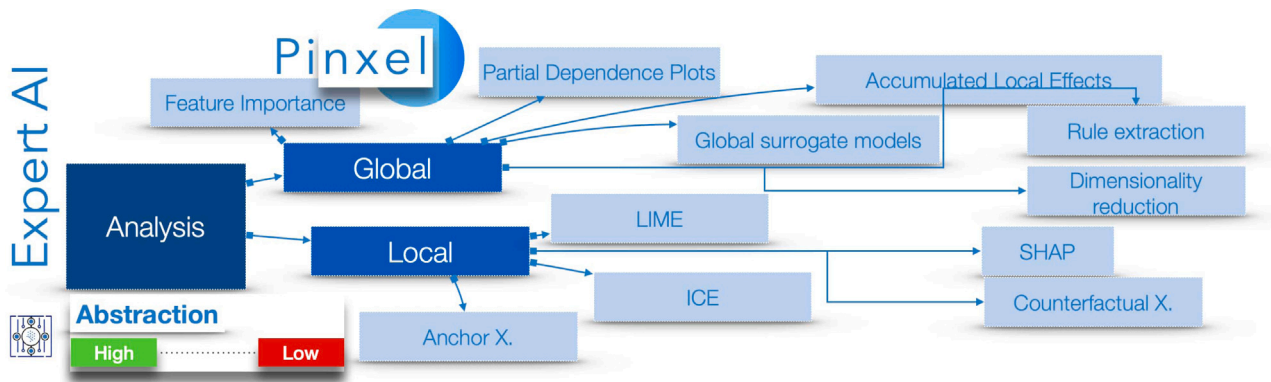


Fig. 5. Expert AI.

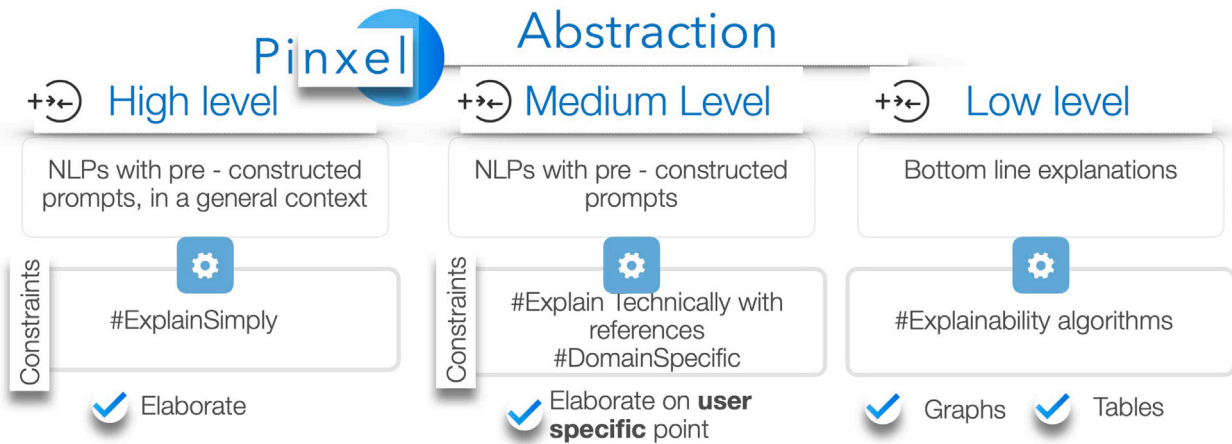


Fig. 6. PINXEL, levels of abstraction.

3.3. Personalisation through clusters and depth of explainability based on REALM

The perceived ease of use and perceived usefulness, alongside the corresponding external variables, namely AI literacy and level of abstraction, allows for the setting of different explainability requirements and transparency layers, thus different explainability depths. However, they may also add unmanageable complexity to the system design. Clearly, the goal should be to maximise usefulness without adding unnecessary burden to a system. Therefore, the question arises as to what the optimum number of clusters is and what the related goals are. A first goal is to improve **perception** of usefulness and ease of use, while at the same time ensure the level of personalisation remains manageable and understandable for the users. Too many clusters might lead to over-personalisation, which could be confusing or overwhelming (Tang, Du, Wang, & Wang, 2016). A second goal is to ensure **performance and efficiency**, by setting a maximum number of clusters, which can limit the required computational resources and, thus, result in more efficient operations (Hastie et al., 2009). In the medical domain, as briefly noted in Section 2.1.2, the Rapid Estimate of Adult Literacy in Medicine implies that the optimal number of clusters is four (4) (Davis et al., 1993). This can be seen in Table 1.

We have taken into account three distinct levels of AI literacy and abstraction. Consequently, we have established three clusters for each of these two levels (AI literacy and abstraction), as suggested by PINXEL. We then employ a questionnaire to identify the various user attributes. To define the clusters, we use ranges based on a Likert scale to assess the questionnaire responses, as detailed in Table 2.

The following formula was constructed as a generalisation of the process.

- For any item x on the Likert scale, $x \in L$, where $L = \{0, 1, 2, 3, 4, 5\}$
 $\forall x \in L, L = \{0, 1, 2, 3, 4, 5\}$
- We have a set of questions $Q = \{q_1, q_2, q_3, \dots, q_n\}$
 $Q = \{q_1, q_2, q_3, \dots, q_n\}$
- In the n -dimensional Likert space, $R = (r_1, r_2, r_3, \dots, r_n)$, where each r_i corresponds to a response for question q_i
 $R = (r_1, r_2, r_3, \dots, r_n)$
 with
 $r_i \in L$ for all $i \in \{1, \dots, n\}$
- We have K clusters, C_k (where k ranges from 1 to K)
 $\{C_1, C_2, \dots, C_K\}$
- Maximum clusters are set to 3, thus
 $K_{max} = 3$
- Vector R is assigned to cluster C_1 if its response vectors are more similar to those in C_1 than to those in any other cluster C_k , according to a specified clustering method such as K-means, Hierarchical, DBSCAN, etc.
 $C_k = \{R \mid R = (r_1, r_2, \dots, r_n),$
 $r_i \in L \forall i \in \{1, \dots, n\},$
 and R is more similar to the representative vector of C_k (e.g., the centroid) than to the representative vectors of the other clusters}

Table 1
Rapid estimate of adult literacy in medicine.

Raw Score	Grade Range
3rd Grade and Below	More effectively understands content presented through illustrations, audio, or video formats
4th to 6th Grade	low literacy levels, may struggle with reading prescription labels.
7th to 8th Grade	Difficulty with most patient education materials
High School	ability to read most patient education materials

Table 2
AI literacy and level of abstraction total clusters.

Clusters	Likert Ranges	Abstraction	Literacy	Explainability depth
1	(0)–2.6	High Level Abstraction (HLA)(1)	Basic AI literacy (BAIL)(*1)	1
2	2.6–3.8	Medium Level Abstraction (MLA)(2)	Average AI literacy (AAIL)(*2)	4
3	3.8–5	Low Level Abstraction (LLA)(3)	Expert AI literacy (EAIL)(*3)	9

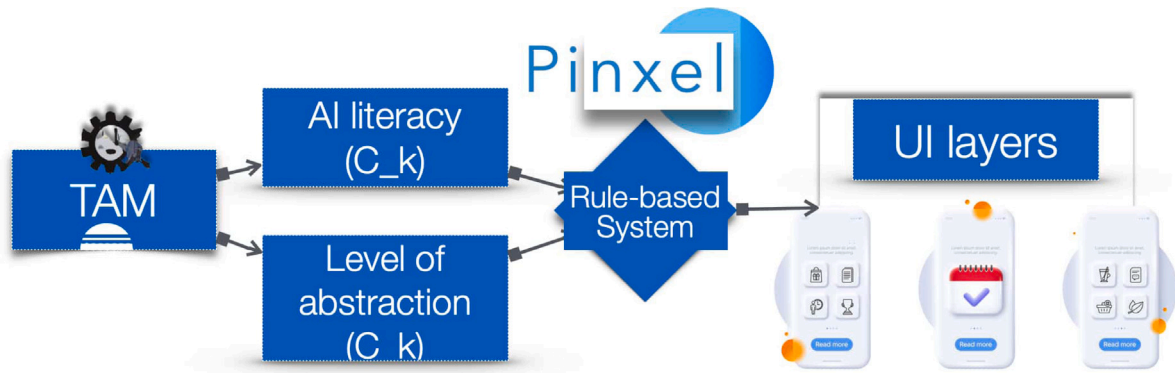


Fig. 7. Analysing the requirements and setting the transparency layers.

3.4. Rule based personalisation

PINXEL allows us to associate perceived usefulness with AI literacy levels and perceived ease of use with required levels of abstraction. As can be seen in Fig. 7, a rule-based system based that considers the TAM defined levels is applied. In this phase the required depth of explainability that is related to the aforementioned levels (AI literacy and level of abstraction) is determined. The following are the corresponding levels of AI literacy and levels of abstraction:

'Basic AI literacy' (BAIL)(1) 'Average AI literacy' (AAIL)(2)

'Expert AI literacy' (EAIL)(3) 'High Level Abstraction' (HLA)(1)

'Medium Level Abstraction' (MLA)(2) 'Low Level Abstraction' (LLA)(3)

Using those levels, the set of rules are the following:

- IF BAIL(1) and HLA(3) THEN Explainability Depth is 3 USE Corresponding levels
- IF EAIL(3) and LLA(3) THEN Explainability Depth is 9 USE Corresponding levels
- IF EAIL(3) and HLA(1) THEN Explainability Depth is 3 USE Corresponding levels

• ...

In the first example, if a person has a Basic AI literacy and requires a High level of abstraction, then the system would provide a corresponding explanation layer. That layer combines the themes shown in Fig. 3 and the tools and constraints that correspond in high level, shown in Fig. 6. Similarly, all the produced rules will lead to a unique output. The max set of rules depend on the total clusters calculated per use case.

4. PINXEL — personalisation analysis for NuhealthSoft

To extract the levels of XAI and define the corresponding depth of XAI (TAM), a distributed questionnaire was constructed, distributed and analysed. Our survey was conducted among 40 students of medicine and practicing doctors, out of which 66.7% were men and 33.3% were women of different specialisation, professional experience and sector of employment. Of the 40 doctors, 39 filled in the form and 1 declined to do so. The electronic questionnaire was anonymously filled in and no personal data were collected. To ascertain the reliability and quality of the results, more than 70% of the participants completed the form under supervision and with the assistance of a person involved in the research.

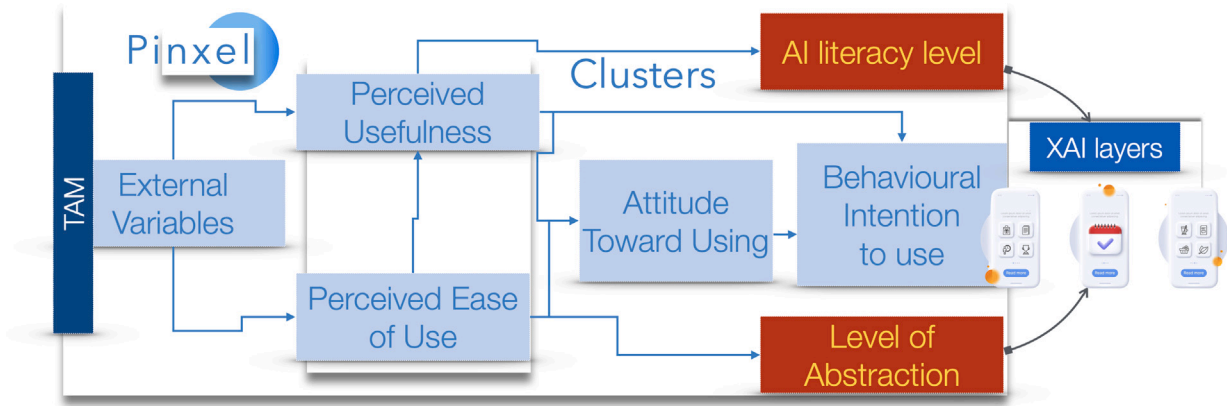


Fig. 8. Adapting TAM for PINXEL.

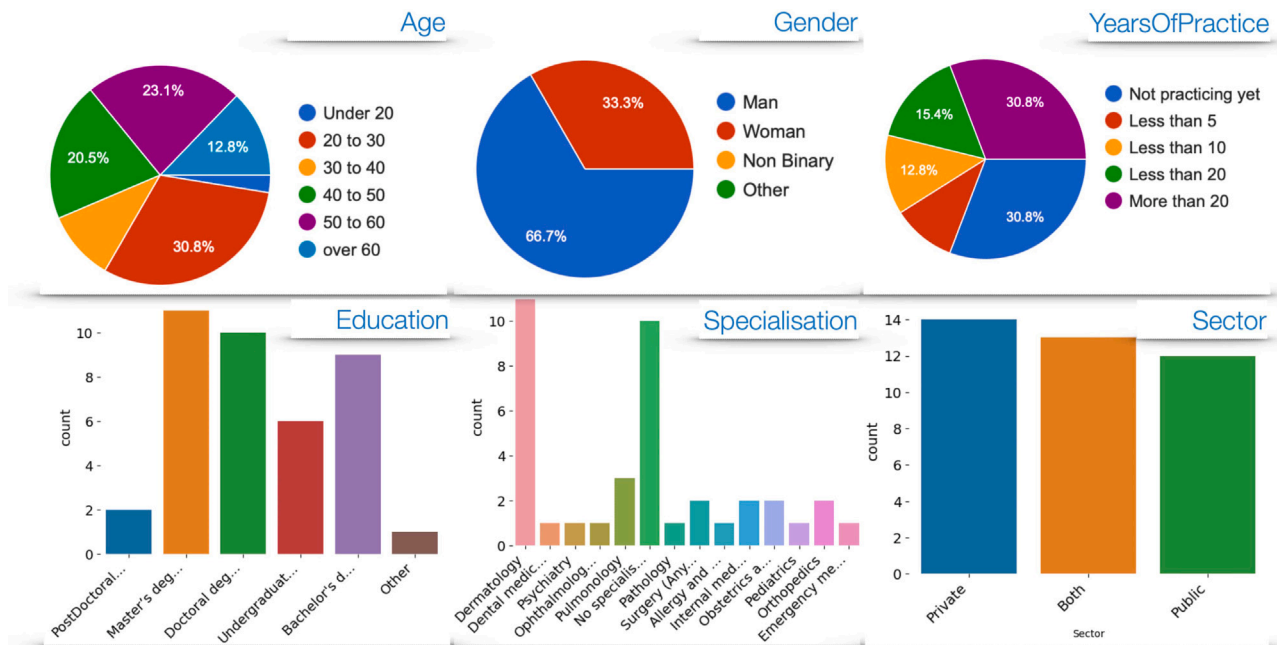


Fig. 9. Demographics.

4.1. Demographics

In Fig. 9 the basic part of the first section of the questionnaire is displayed. The bulk of the respondents are over 40 years old, with a minor faction of 10 being undergraduate students under 30 years old. Predominantly, the respondents have ties to hospitals, with a near-even split of clinicians working in both public and private sectors. The respondents have a diverse range of specialties, spanning across various aspects of the medical field as depicted in Fig. 9. The scale of practice also exhibited diversity, assessed via a Likert scale in which 0 denoted no employees in the practice and 5 indicated more than 10. Fourteen (14) respondents were employed in private practices with a staff of 2 or more, whereas 6 worked in establishments having a workforce exceeding 10. A fraction of the respondents (namely 5) held the role of hospital unit directors, while 2 of them were academicians in a University. The pinnacle of educational attainment is the post-doctoral level, attained by 2 survey participants. Ten respondents held a doctorate degree, while 11 held a master's degree. The respondents' proficiency in English was assessed and rated to be superior than average. Their professional experience varies, ranging between less and more than 10 years, with around 30% having been in the medical field for over two decades.

4.2. PINXEL — perceived ease of use (TAM) and levels of abstraction

The questions that quantify perceived ease of use (Table 3), were constructed taking into account a series of parameters that would facilitate the measurement of the required level of abstraction and by association, the participants' perceived ease of use of an AI-empowered system. We theorised that this group of questions give a sense of the participants' attitude towards innovation and technology, as well as their attitude towards novel approaches to medicine like tele-health. Perceived ease of use refers to the extent to which a person believes that employing a particular technology will be straightforward and devoid of major complications. Specifically, questions Q1 and Q4 aim at assessing the general disposition towards therapeutic and pharmacological advances in medicine, due to the reasons outlined previously. The remaining questions provide insight into how physicians utilise and perceive contemporary innovations in their field. By characterising these diverse attitudes, specific measures could be implemented by proffering a customised experience based on the identified attitudes. This method would enhance the perceived ease of use and, thereby, boost the acceptance and integration of the associated technology. This can be achieved by highlighting the AI properties of the system in a manner tailored to individual characteristics and abilities.

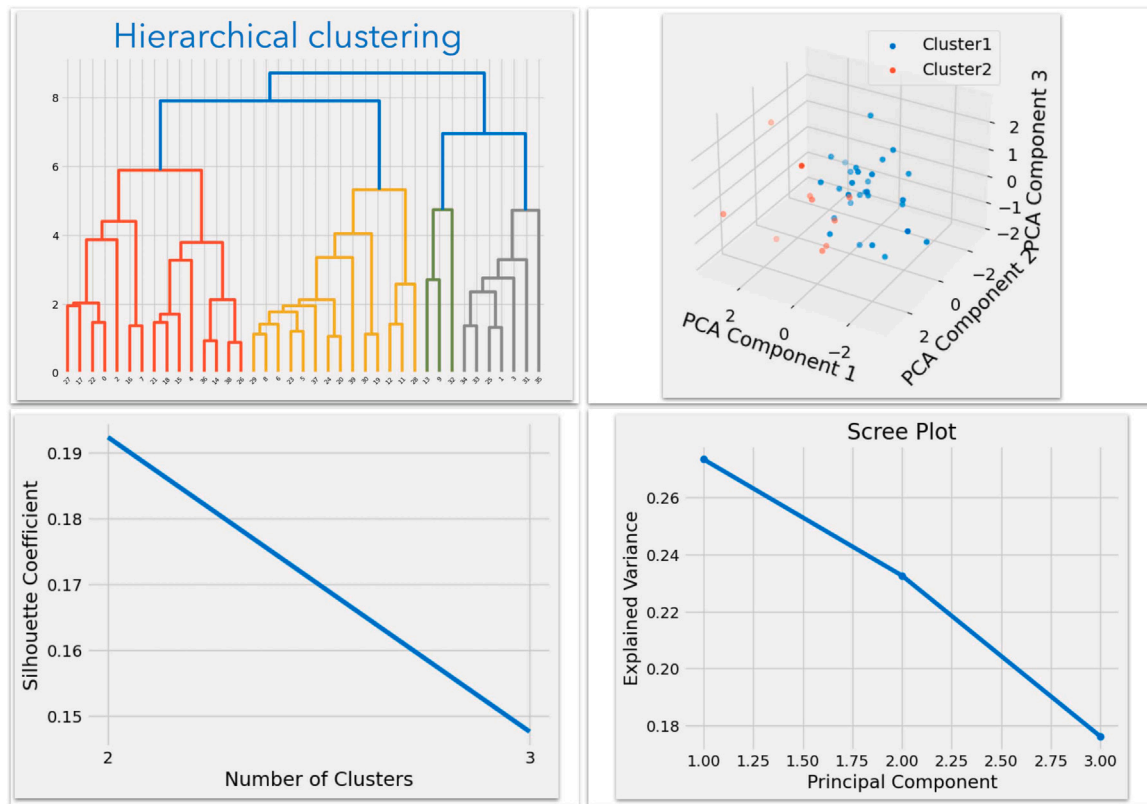


Fig. 10. Perceived ease of use and levels of abstraction — clusters.

Table 3
Perceived ease of use and associated level of abstraction, survey questions.

Question	Perceived ease of use
Q1	If a new therapy or medicine is introduced, how likely is it that you propose it or will propose it to a patient or use it on yourself, early in its introduction phase
Q2	Does the medical equipment that you use, have automation to provide suggestions per treatment or expected outcome
Q3	Telehealth is the distribution of health-related services and information via electronic information and telecommunication technologies. Do you feel that, at least in some cases, it can provide outcomes similar to in person visits
Q4	Do you find that extensive clinical trials of new technologies or medicine are a lengthy process or the rule 'safety first' needs to be followed
Q5	Do you feel that patients searching for symptoms online is beneficial or problematic?
Q6	Is human-to-human communication more important when a health issue is more severe or is it unrelated to the severity of a health issue?

Hierarchical clustering was applied on the questions from Table 3, initially using silhouette coefficients to determine the optimal number of clusters. The outcomes are presented in Fig. 10. Based on our examination, six distinct clusters are associated with the participants' respective AI literacy levels. However, upon further scrutiny, it was discovered that one participant selected 0 in response to all the questions, leading to the final evaluation of only five clusters. The key attributes, derived from the average score of each response, are detailed in Table 4. To illustrate the various clusters, we utilised three components which, as indicated in the principal component analysis of Table 4, account for more than 50% of the total questions (6).

Table 4
Clusters based on perceived ease of use — Level of abstraction.

Question	$C_1(\text{mean})$	$C_2(\text{mean})$	Q(Average)
Q1	3.48	3.5	3.49
Q2	2.24	1.4	1.82
Q3	3.31	3.0	3.155
Q4	4.48	2.8	3.64
Q5	1.72	2.9	2.31
Q6	2.24	3.2	2.72
TotalCount	29	10	
Average	2.93	2.8	

Table 4 elucidates the distinctive scores for each question and cluster. The question receiving the lowest score was question Q5, implying a predominant sentiment among participants that *patients' online investigation of symptoms poses potential issues*. In relation to question Q2, the study data indicates that *the participants have not used medical devices with implemented automations*, such as those providing treatment recommendations or predictive outcomes. Conversely, question Q4 received the highest score, hinting at an *inclination of physicians towards prioritising safety* in the context of conducting clinical trials for novel technologies and diagnostics or assistive technologies for medical diagnosis.

Using the score system proposed in Table 2, we identify 2 clusters but 1 level of required abstraction. It should be noted, that by having a deeper level of understanding of the users and their individual characteristics, additional levels of abstraction could be engineered in the future, thus a deeper XAI to be constructed. In this case, the required level of abstraction is 'Medium Level Abstraction' (MLA)–2.

4.3. PINXEL — perceived usefulness (TAM) and AI literacy levels

As has been proposed and established in this paper, a strong association exists between technology literacy and perceived usefulness,

Table 5
Clusters of perceived usefulness — AI literacy.

Question	C_1 (mean)	C_2 (mean)	Q(Average)
Q1	3.40	3.0	3.20
Q2	3.33	3.25	3.29
Q3	3.96	2.91	3.70
Q4	4.11	3.25	3.435
Q5	4.55	2.75	3.65
TotalCount	27	12	
Average	3.87	3.03	

Table 6
Perceived usefulness and associated AI literacy, survey questions.

Question	Perceived usefulness
Q1	Can you indicate your level of knowledge on diagnostic medicine
Q2	Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world. Would you trust it as a feature in driving automation
Q3	AI can add value by either automating, assisting or augmenting the work of clinicians and staff. Many repetitive tasks will become fully automated, and we can also use AI as a tool to help health professionals perform better at their jobs and improve outcomes for patients. How much do you agree with the above statement
Q4	If using AI-empowered systems do you feel that the more you know about the underlying technology the better, or you had trust the provider
Q5	Would you participate in developing AI medical applications?

reinforcing our approach to employ AI literacy as a metric for this external variable. Table 6 illustrates the selected questions that estimate perceived usefulness and corresponding AI literacy. Q1 gauges the participants' understanding of diagnostic medicine, a domain that employs statistical methods and data analysis to enhance diagnosis and treatment decisions. The participants' grasp of diagnostic medicine suggests a comprehension of statistics and data analysis, thus hinting at an elevated AI literacy level. Q2 aims at revealing the degree of understanding and trust in computer vision, particularly when employed in autonomous driving, a widely commercialised application of this technology. The assumption is that increased trust corresponds to a more profound understanding and a higher AI literacy level in respondents. Similarly, Q3 and Q4 outline fundamental functionalities of AI, thereby reflecting the respondents' AI comprehension and literacy. Finally Q5, queries on the willingness of the participants to take part in developing AI-empowered medical systems. The scale of the response indicates a higher perception of usefulness, with regard to this particular technology.

Employing the same approach, hierarchical clustering is utilised. As per the silhouette coefficients, two clusters are identified as optimal (see Fig. 11). These clusters can be observed in Fig. 11, where Principal Component Analysis (PCA) has been employed to reduce the dimensions of the space of features-questions from 5 to 3. Question Q2, which assesses the trust in computer vision as a driver for automated driving, scores the lowest. In contrast, question Q5 scores the highest, illustrating an elevated inclination for involvement in the evolution of AI-empowered medical systems.

Using the score system proposed in Table 5, we identify 2 levels of AI literacy.

- Expert AI literacy level, clusters: C1
- Average AI literacy level, clusters: C2

By utilising the rule-based system described previously, the implementation strategy, which hinges on the clustering delineated by the two levels of PINXEL, will proceed as outlined:

- (C1) IF EAIL(3) and MLA(2) THEN Explainability Depth is 6 USE Corresponding levels
- (C2) IF AAIL(2) and MLA(2) THEN Explainability Depth is 4 USE Corresponding levels

In summary, the subjects outlined in Figs. 4 and 5, are to be explained, using a Medium level of abstraction, i.e more technical questions to be answered by NLPs empowered by Large language models (LLMs), as proposed in Fig. 6.

5. Implementation examples in NuhealthSoft

In Fig. 12, an outline of the layers of the system are presented. The first layer contains the database (D.b) controllers, which includes the business logic of the system and the ML algorithms that provide classification and recommendation methods. The second layer contains the structural semantics, i.e. the inputs, outputs, stakeholders and associated context, which are extracted by the D.b controllers and provide the semantic groupings as XAI objects. Also in this second layer, the json summaries of the different ML parameters are stored, again as objects to be utilised in user interfaces where different AI literacy levels and levels of abstraction are taken into account, as outlined in the previous sections. These user interfaces are defined in the final (third) layer of the system, which contains the proposed transparency and XAI framework, as described previously, and implemented in accordance with the pre-set explainability depth (Expert, average, basic etc.).

5.1. Addressing concerns, towards building responsible systems

All the different parameters discussed in this paper, encompass what can generally be perceived as a strategy towards transparency, interpretability and inclusiveness. In the following subsections, some different use cases are provided in accordance to PINXEL, and implemented in NuhealthSoft.

5.1.1. PINXEL — average AI literacy level with low level of abstraction

In Fig. 13, an implementation example can be seen for an average literacy level with Low level of abstraction. The architecture part of this figure demonstrates the different layers of the trained model. It consists of an input layer of 18 neurons in total. The input layer receives the raw data in the form of feature vectors. The output is a transformed version of the input layer which is passed to next layer, if one exists. The second layer, named “dropout,” is a regularisation technique that prevents over-fitting by randomly dropping out neurons during training. In this case, 50% of the neurons from the previous layer are dropped and fed to next layer. The last dense layer extracts a binary prediction or classification. The parameters tab refers to the training parameters of the neural network, i.e. the weights and the biases. These parameters break down the symmetry of the fed input to ensure that different neurons learn different features. Then the weighted sum of inputs of each neuron is calculated using an activation function.

Using optimisation techniques, loss calculation and continuously updating the parameters, the weights and biases change towards the optimum result, i.e. towards a prediction or classification both on training and testing data. In this case the parameters change for each layer, since a dropout layer is being applied. The formula to calculate the total parameters for each layer can be seen below: parameters = (input neurons + 1) * output neurons. The +1 accounts for the bias associated with each output neuron.

In the same figure, we can see the distribution of the weights and biases of the first layer. The confusion matrix shows the relevant accuracy of the final trained model on the testing data. Finally, the history plots show how the training-validation loss and accuracy change over time, in the case of training a neural network over a series of training epochs.

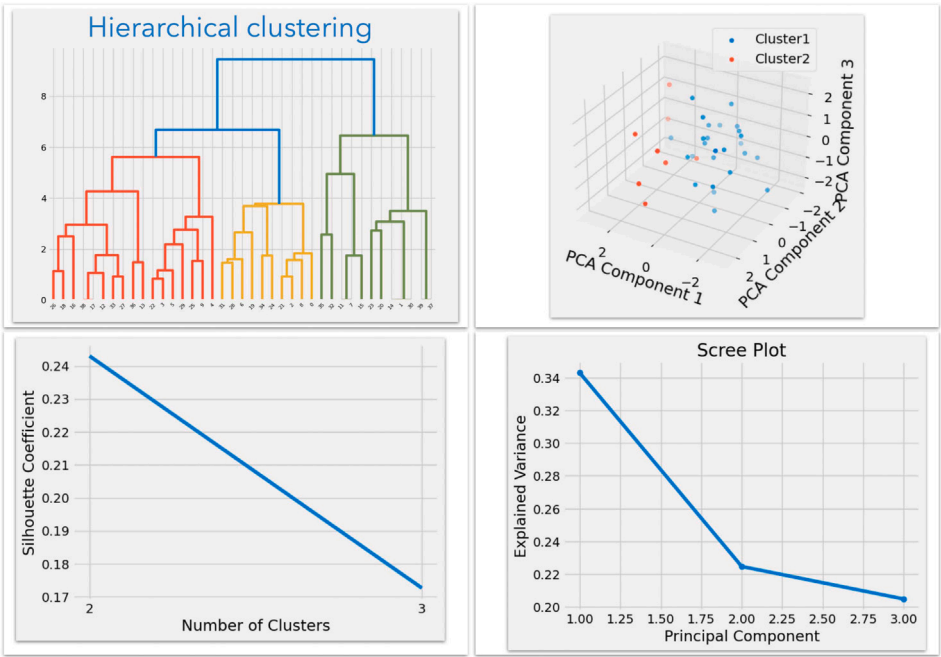


Fig. 11. AI literacy — Perceived usefulness.



Fig. 12. XAI as feature.

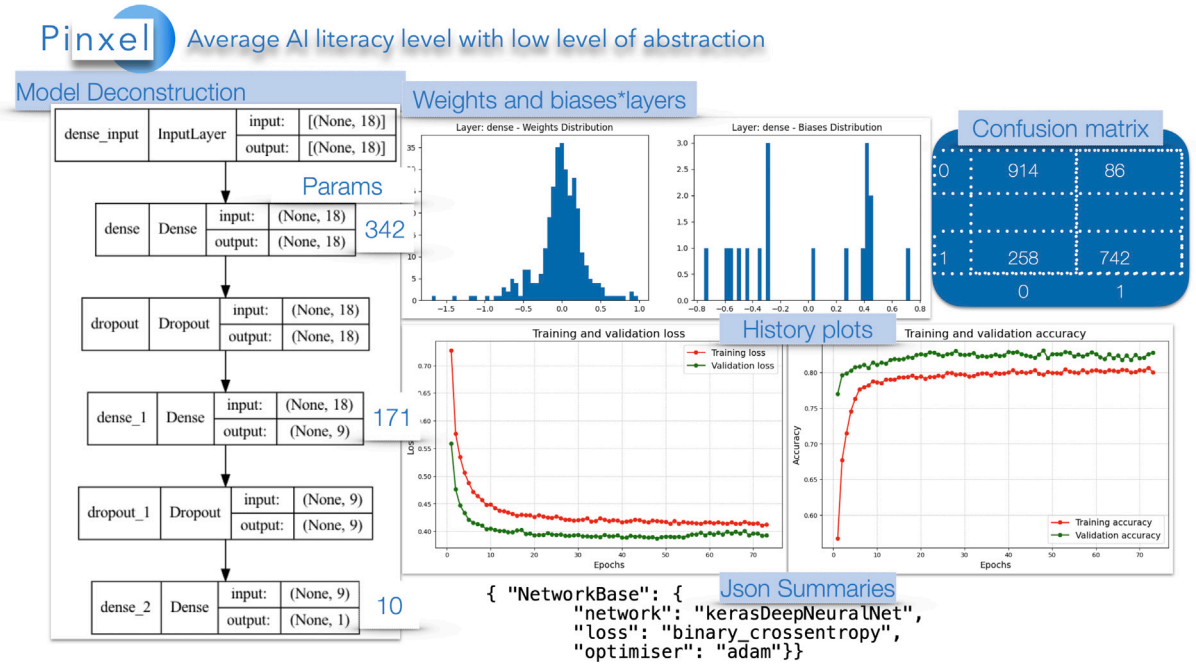


Fig. 13. Model deconstruction — Model evaluators.

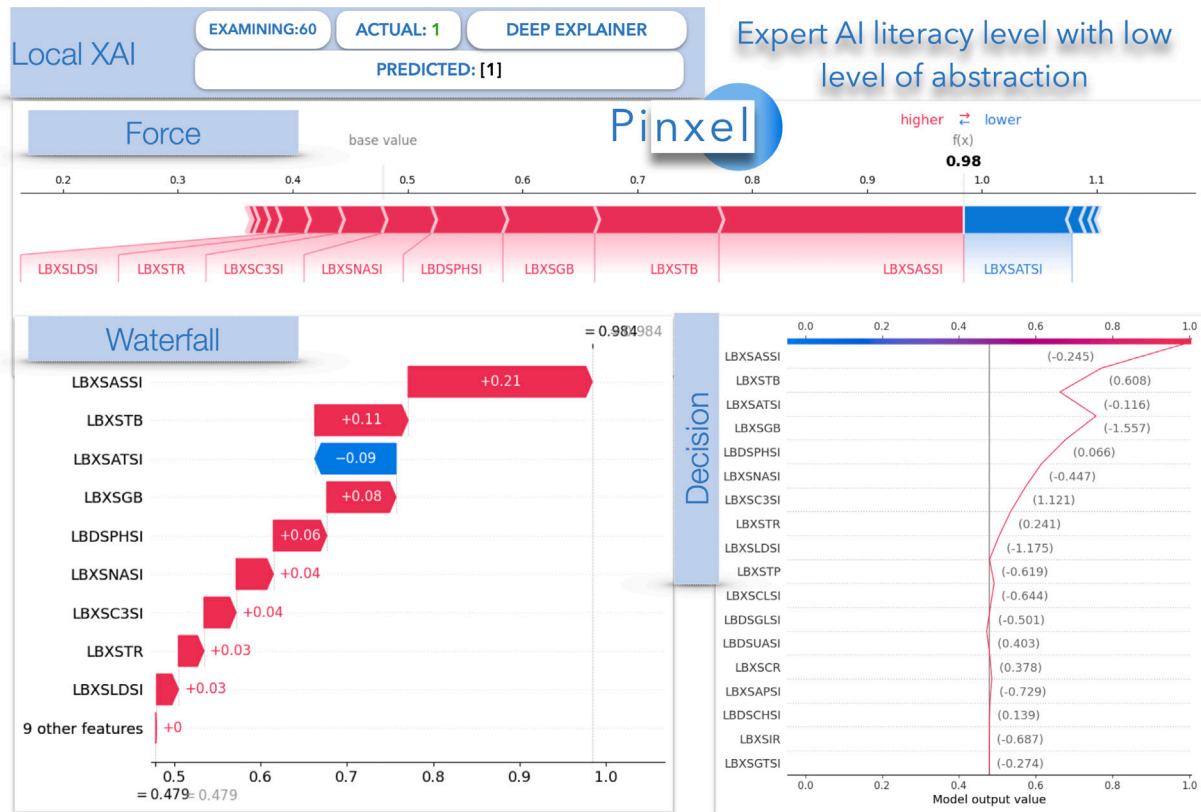


Fig. 14. Local XAI — NuhealthSoft weight based on blood exams.

5.1.2. PINXEL — high AI literacy level and low level of abstraction

In Fig. 14, an implementation example for High AI literacy level with Low level of abstraction can be seen which Shapley values are used. We have focused on datapoint 60 (out of approximately 10,000 datapoints included in the training sample) to understand how various features influenced the model prediction of “1”, which concurs with the actual value. More specifically, an output of “1” signifies the overweight and obese weight category, whereas an output of “0” represents the underweight and normal weight category. The employed force plots elucidate how each individual feature – illustrated through a blood variable in this case – influenced the final decision made by the trained model. These plots delineate how each feature either pushed the prediction towards or pulled it away from the base value, thereby representing the average probability of a prediction.

Waterfall plots, akin to force plots, visualise the step-by-step contribution of each feature towards the final prediction. However, these plots underscore the incremental impact each feature has on the final decision. Lastly, decision plot compile the cumulative effects of all the features, demonstrating how they collectively influence the final output value or prediction.

Again, the Shapley values are used to depict the global XAI (Fig. 15) parameters under the same PINXEL depth of explainability. In the feature importance plot, the average contribution of each feature is calculated for the model’s overall predictive performance (Lundberg & Lee, 2017). Here, the list of each blood variable can be seen from the full list of variables in Panagoulas et al. (2021b)). Specifically, the Partial Dependence Plots show the relationship between the actual value of a single feature (x_axis) and the shap values of this feature (y_axis), as extracted from the ML model. The colours of the dots represent the value of another feature, thus visualising the interaction between features.

5.1.3. PINXEL — basic AI literacy level with low level of abstraction

In Fig. 16, we present a sequence of user interface screenshots illustrating a typical instance of semantic grouping.

In this particular instance, Chloride, with its associated coded name and unit of measurement, has been selected for examination. The second screen displays detailed information linked to this blood variable, a crucial component of the standard biochemistry profile and an integral part of the NuhealthSoft ML pipeline. This screen also visually articulates the relationship between chloride levels and specific organ functions. Indeed, the kidneys and heart are shown to be influenced by chloride. Subsequently, the kidneys are selected for further exploration.

In the third (final) screen, comprehensive information related to the kidneys is displayed. Towards the bottom of this screen, we can see a depiction of the interrelationships between the kidneys and other blood variables that form a part of the standard biochemistry profile. This visualisation aids the understanding of the complex network of interactions within the body biochemistry and, thus, the way a ML model can identify patterns and create classification rules.

5.1.4. PINXEL — low AI literacy level with high level of abstraction

In Fig. 17, we present a chat bot used in NuhealthSoft. This is a personalised screen for low AI literacy level users employing the simplified logic of higher level of abstraction. The predefined prompts are structured as buttons taken from the corresponding low AI literacy level story line layers and all the associated themes used in that part of the explainability pipeline. As can be seen in Fig. 17, in order to navigate the basic thematic concepts, the user is provided with the related choices related to ML. Having achieved a basic understanding of these themes, the user can advance his understanding and gradually get acquainted with more technical concepts and application parameters.

6. Summary of results and future works

In this research paper, we introduce a novel framework called PINXEL, designed to address the challenge of personalisation in Explainable Artificial Intelligence (XAI). Our framework takes into account users’ AI literacy levels and levels of XAI abstraction, enabling

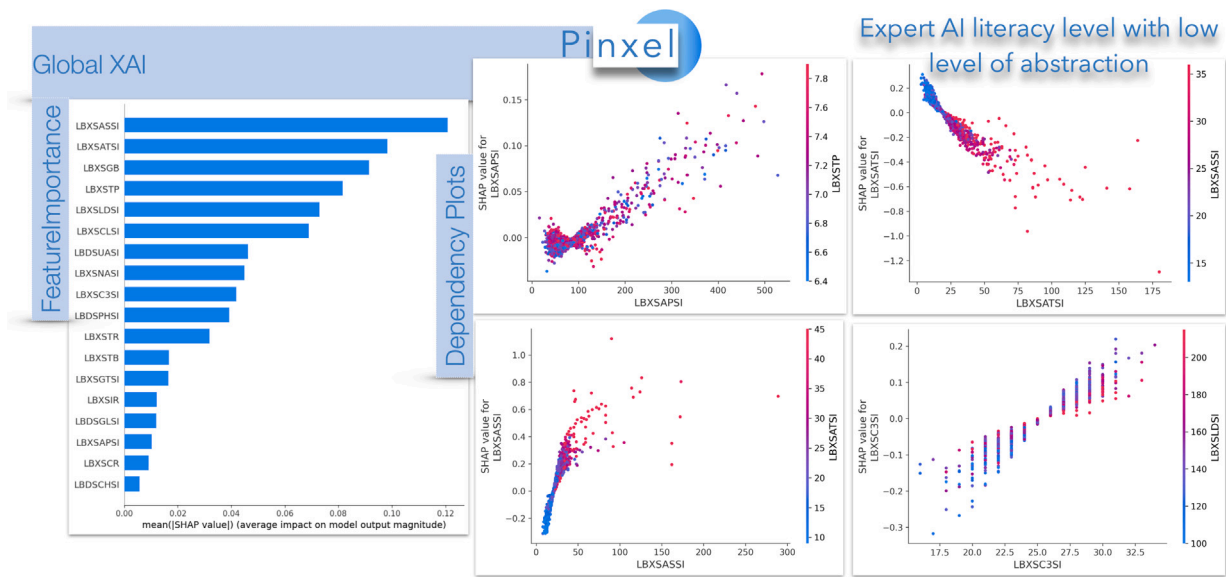


Fig. 15. Global XAI — NuhealthSoft weight based on blood exams.

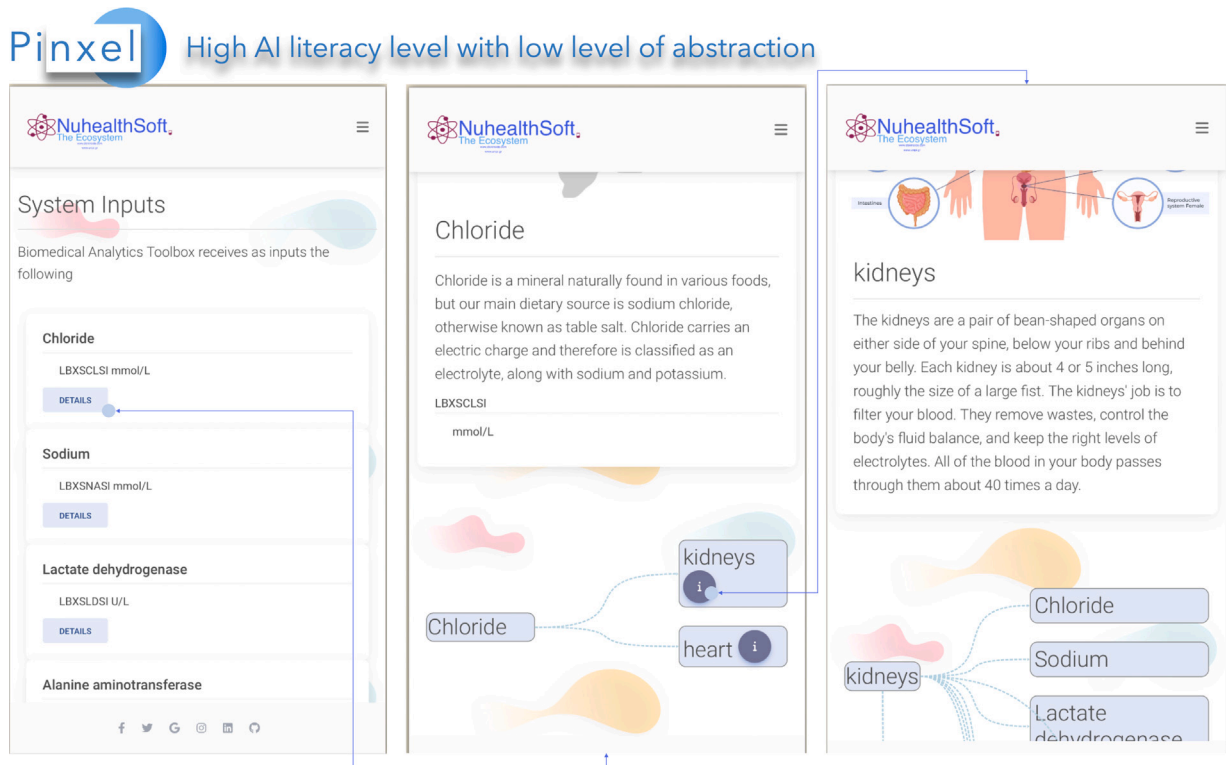


Fig. 16. Semantic groupings — NuhealthSoft weight classification, based on blood exams.

the application of clustering techniques to questionnaires aimed at assessing users' abilities and knowledge in the domain of AI systems. By incorporating XAI user interface (UI) layers, especially in sensitive domains, we aim to facilitate users in validating, comprehending, and ultimately embracing AI-empowered systems.

The implementation of PINXEL involves the use of a rule-based system to offer personalised layers of XAI based on the TAM. XAI themes are grouped based on complexity (AI literacy). The abstraction problem is addressed using Large language models, graphical representations and tables, to discuss the various (grouped) XAI themes.

To evaluate the efficacy of our framework, we conducted a use case analysis by examining questionnaire responses based on PINXEL

prerequisites. Furthermore, we analysed a specific use case pertaining to the integration of different XAI layers in an AI-empowered medical application called NuhealthSoft.

In our future research endeavors, we are committed to rigorously validating our system and processes. This will be achieved through A-B testing to provide clear comparisons between scenarios A (AI without XAI) and B (XAI with PINXEL). Our objective is to quantitatively measure the impact of our framework on the adoption and satisfaction rates of XAI-empowered systems. Furthermore, we are expanding our methodology to encompass the evaluation of patient requirements using the PINXEL framework, which is particularly pertinent in medical applications.

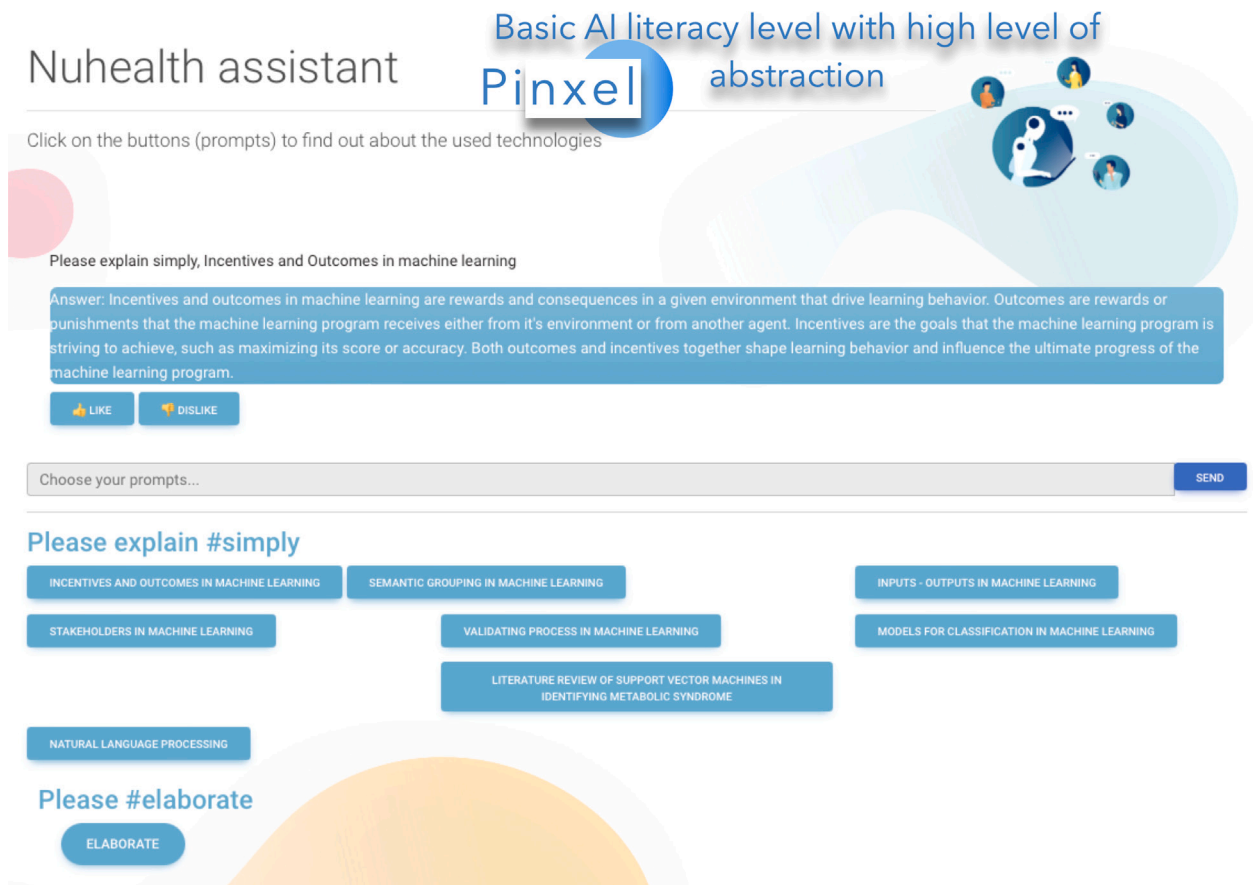


Fig. 17. PINXEL — Low AI literacy levels with High level of abstraction.

Addressing the main limitations identified in our study forms a crucial part of our future work. Our initial focus has been on a specific domain, where the educational background and abilities of users are more homogeneous. However, the diversity found in the general population presents a considerable challenge. To effectively cater to this diversity, we recognise the need to incorporate various behavioural models and theories. This approach will help us in making our system more universally applicable and user-friendlier.

A pivotal aspect of our research is the privacy and security of user data, especially in the context of medical information used in our machine learning and analytical services. Although we have partially addressed these concerns in our previous work, we acknowledge the need for continual improvement. Our future goal is to explore more intuitive methods, such as federated learning and blockchain technologies, to enhance data security and user privacy.

Another significant challenge we face is the resource intensity required for maintaining AI systems with integrated XAI tools. This challenge is amplified in an era where technological advancements are rapid and constant, leading to frequent updates and improvements. Our future focus will be on investigating decentralised systems that boast improved inter-connectivity. Such systems have the potential to facilitate and minimise efforts associated with maintenance, while also efficiently managing and monitoring the costs of related resources. By tackling these challenges, we aim to make our framework more robust, adaptable, and user-centric.

Abbreviations

The following abbreviations are used in this manuscript:

ML	Machine learning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
NLP	Natural Language Processing
LLM	Large language model
TAM	Technology Acceptance Model
RAI	Responsible Artificial Intelligence
REALM	Rapid Estimate of Adult Literacy in Medicine
TOFHLA	Test of Functional Health Literacy in Adults
NVS	Newest Vital Sign
PDP	Partial Dependence Plots
ALE	Accumulated Local Effect
LIME	Local Interpretable Model-Agnostic Explanations
SHAP	SHapley Additive exPlanations
ICE	Individual Conditional Expectation
BAIL	Basic AI literacy
AAIL	Average AI literacy
EAIL	Expert AI literacy
HLA	High level of abstraction
MLA	Medium level of abstraction
LLA	Low level of abstraction

CRedit authorship contribution statement

Dimitrios P. Panagoulas: Conceptualization, Methodology, Software, Writing – original draft, Visualization. **Maria Virvou:** Conceptualization, Writing – review & editing, Supervision, Methodology. **George A. Tsihrintzis:** Conceptualization, Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data analysed in this study can be found in csv format at the following link: <https://www.diskinside.com/wp-content/uploads/2023/12/A.I.-in-medicine.csv>.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.eswa.2024.123375>.

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